HAOYU WANG

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Profile

PhD candidate in Financial Engineering at the University of Auckland. Research interests include machine-learning asset pricing, factor discovery, portfolio optimization, market microstructure and reinforcement learning. Develops LLM-assisted tools for quantitative research, data analysis and reporting.

Education

University of Auckland Dec 2024 - Present

PhD, Financial Engineering - Full Scholarship | Auckland, New Zealand

Xi'an Jiaotong-Liverpool University Sep 2022 - Apr 2024

MSc, Financial Mathematics

Chongqing Finance and Economics University Sep 2018 - Jun 2022

BSc, Finance

Publications

Optimizing Stock Market Prediction and Stock Trading Strategies with Deep Learning Models Enhanced by Nonlinear Feature Identification and Robust Prediction Evaluation 2026

Financial Innovation, 12, 117. DOI: 10.1186/s40854-026-00929-6

Information-aware Valuation and Dynamic Pricing for Textual Data in Mobile Environments 2026

Information Processing and Management, 63 (7), 104855. DOI: 10.1016/j.ipm.2026.104855

Can Ensemble Learning Reveal Real Estate Value – A Study of England and Wales Cases Anonymous 2026

with Yinze Ji and Yishuang Xu. *Journal of Chinese Economic and Business Studies*, 1–36. Routledge. DOI: 10.1080/14765284.2026.2722767

Optimal Profit-Making Strategies in Stock Market with Algorithmic Trading 2024

Quantitative Finance and Economics, 8 (3), 546. American Institute of Mathematical Sciences (AIMS).

Working Papers

Too Much of a Good Thing: Characteristic Redundancy and Machine-Learning Portfolio Performance 2026

With D. Jayasuriya, L. Xiao and X. Han. Working paper; AFBC 2026 main program.

Entropy-Regularized Portfolio Optimization with Reinforcement Learning Control 2025

Working paper. SSRN 5276183.

AI & Automation

LLMs (Claude, Codex), prompt engineering and AI-assisted development for data cleaning, reconciliation, reporting, model prototyping and repeatable research pipelines.

Additional Teaching

Xi'an Jiaotong-Liverpool University 2023

Teaching Assistant, Calculus

Presentations & Competitions

39th Australasian Finance and Banking Conference (AFBC) Dec 2026

UNSW Sydney - Main program (upcoming)

Citadel Trading Invitational 2025

Australia

Market-making simulation, quant simulation and timed coding challenge.

Awards & Certifications

XJTLU Entrance Scholarship (2022) · CMA Part 2 passed: Strategic Financial Management · Securities Practitioner Qualifications, 3 modules (2019) · NCRE Level II (2019).

Research & Industry Experience

University of Auckland Dec 2024 - Present

Financial Researcher | Auckland, New Zealand

Alpha discovery, factor mining and portfolio optimization; LLM-based automated research and strategy generation; reinforcement learning for market microstructure.

University of Auckland Jul 2025 - Present

Graduate Teaching Assistant, Fintech

Oxford Finance Limited Oct 2025 - Dec 2025

Finance Analyst (Contract) | Auckland, New Zealand

Financial data analysis.

Shenzhen Research Institute, The Chinese University of Hong Kong Jan 2024 - Dec 2024

Research Assistant (Contract) | Shenzhen, China

Guohua Life Insurance Co., Ltd. Mar 2024 - May 2024

Actuarial Intern

Asset-liability management.

S&P Global Oct 2023 - Feb 2024

Quantitative Research Intern, IHS Automotive Planning Solution | Shanghai, China

Machine-learning pricing analysis of BEV, MHEV and PHEV models; published the 2023 China NEV Export Outlook using data-driven modeling and market forecasts.

Xi'an Jiaotong-Liverpool University Oct 2022 - Feb 2024

Research Assistant, School of Mathematics and Physics | Suzhou, China

Calculus, stochastic processes and American option pricing; machine learning and data analysis.

Kaiyuan Securities Aug 2023 - Oct 2023

Quantitative Intern

Volvo Group Mar 2023 - Aug 2023

Data Analyst Intern | Shanghai, China

Market-dynamics and competitive-landscape reports for the sales department.

Selected Projects & Research

FactorForge - LLM Factor Research

LLM-guided factor generation, predictive diagnostics and out-of-sample portfolio research across US equities, China A-shares and crypto perpetuals.

Hybrid RL Market Making with Avellaneda-Stoikov Backbone

Reinforcement-learning market-making research with an Avellaneda-Stoikov baseline.

Avellaneda-Stoikov Market Making

Market-making and inventory-control research using Binance Futures BTCUSDT order-book data.

Limit Order Book Replay

Order-book replay tools for market-microstructure research.

A-share Alpha Factor Lab

Alpha158 and GTJA191 factor research with walk-forward LightGBM evaluation and Tushare data.

Momentum Is All You Need

Equity momentum research covering skip-month signals, volatility targeting and capacity analysis.

NZ Electricity Market Spot-Price Forecasting

End-to-end forecasting of half-hourly NZEM spot prices using Electricity Authority/EMI data.